The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)

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NHS Foundation Trust

Management of Sodium and Water Balance
HYPONATRAEMIA (Na+ <135mmol/L)

CONSIDER TREATING SERUM SODIUM ≤130mmol/L

Normal plasma sodium =135–145mmo/L
& plasma osmolality 285–295mOsm/kg

Sodium disturbances are common after brain injury and treatment is indicated when there are acute symptoms that pose a risk of significant neurological complications.

The symptoms and signs relate not only to the severity of the sodium disturbance but also to the rate of change of serum sodium concentration.

HYPONATRAEMIA is defined as a serum sodium concentration of <135mmol/L and It usually develops between 2 and 7 days after injury and is associated with mortality increases of up to 60%. Hyponatraemia creates a gradient across the blood–brain barrier allowing entry of water into the brain and the development of cerebral oedema. The cause of the low serum sodium can be determined by the associated volume status.

Both **CSW** and **SIADH** can occur following brain injury. It is important to distinguish between the two as the treatment differs. For CSW the patient is given fluids and sodium supplementation, while for SIADH the patient is fluid restricted unless the sodium is severe (<125mmol/L) and the patient has significant symptoms.

CSW is mostly seen in SAH and the patient must be volume repleted while treating the hyponatraemia – with isotonic saline for mild to moderate cases, hypertonic saline (1.8%) for moderate to severe cases. In some cases the use of fludrocortisone is also advocated.

Following confirmation of SIADH, the aim is to restrict the intake of water. Important drug-related causes include diuretics (particularly thiazides) and anticonvulsant drugs (particularly carbamazepine).

The rate of serum sodium improvement in both CSW and SIADH should be gradual (8-10mmol/24h) in most circumstances to avoid complications associated with rapid osmolar shifts (cerebral oedema and osmotic demyelination syndrome). Severe hyponatraemia (irrespective of the cause) may lead to neurological complications and requires prompt management with intense monitoring.

DIAGNOSTIC EVALUATION

- ☐ Exclude normal or elevated P_{osm}
- ☐ Exclude hypotonic hyponatraemia
- ☐ Check urine $_{osm}$
- ☐ Assess volume status and urine $_{Na+}$

	CSW	SIADH
Serum sodium	Low (<135mmol/L)	Low (<135mmol/L)
Serum osmolality	Normal or increased	Decreased (<280mOsm/kg)
Volume status	Hypovolaemia	Normo- or euvolaemia
Urine sodium	Very high (>40mmol/L)	High (>25mmol/L)
Urine osmolality	High or normal	High (urine osmo > serum osmo)
Haematocrit	High	Normal

CEREBRAL SALT WASTING (CSW)

Na⁺ 125-130mmol/L

The primary treatment of CSW is volume and sodium resuscitation

Aim to **REPLACE** sodium and water with sodium containing **ISOTONIC FLUIDS**

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE (SIADH)

Na⁺ 125-130mmol/L

Aim to **RESTRICT** intake of water

FLUID RESTRICTION dependent on individual patient requirements

Consider risk of fluid restriction in patients with SAH
In SAH patient with low Na+ assume CSW until sure of diagnosis of SIADH because of consequences of fluid restriction

Restriction to 1.5L/24h usually sufficient

Na⁺ <125mmol/L

HYPERTONIC SALINE *(HTS)* **1.8%** for **24-48 HOURS** preferably via central venous line to increase sodium concentration by maximum **8-10mmol in 24 hours** (Hourly rise not >2mmol/L unless Na+ consistently low)

1.8% saline can be given peripherally but IV site must be closely monitored – not recommended if treatment required for >48h

RATE OF INFUSION

- ☐ The rate of infusion should be prescribed for each patient individually
- ☐ Set **hourly / daily target rise** in Na⁺
- ☐ Hypertonic saline usually started at ~ **30-40mL/h**
- ☐ Once daily target achieved hypertonic saline should be stopped

Rapid increase or decrease can cause myelinolysis

FLUDROCORTISONE

If hyponatraemia refractory to standard therapy consider fludrocortisone **50mcg/tds for 4 days**
Thereafter wean every 48h: *50mcg/BD for 48h, then 50mcg/OD for 48h - then stop*
Consider slow sodium tablets

MONITOR

- ☐ CARDIOVASCULAR STATUS
- ☐ FLUID BALANCE
- ☐ SERUM SODIUM
- ☐ RENAL FUNCTION
- ☐ URINE AND SERUM OSMOLALITY

BD urine / serum Na⁺ osmolality if receiving HTS
If Na⁺ <125mmol/L check 2 hourly via blood gas

CLINICAL MANIFESTATIONS

Symptoms	Develops at
Lethargy Headache Nausea / Vomiting Irritability Muscle cramps / weakness	*Develops at;* Na+ 125-130mmol/L when ACUTE Na+ 120-125mmol/L when CHRONIC
Hyporeflexia Drowsiness Confusion	*Develops at*: Na+ 120-125 mmol/L when ACUTE Na+ 110-120 mmol/L when CHRONIC
Seizures Coma Death	Na+ <115mmol/L

Presence and severity related to:
1. Severity of derangement
2. Speed at which derangement develops

OSMOTIC DEMYELINATION SYNDROME

Characterised by damage to regions of brain, most commonly pontine white matter tracts, after rapid correction of metabolic disturbances such as hyponatremia